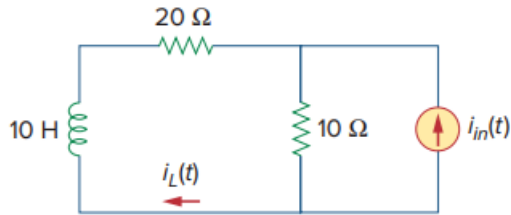


Problem

Determine the value of $i_L(t)$ and the total energy dissipated by the circuit from $t = 0$ sec to $t = \infty$ sec. The value of $i_{in}(t)$ is equal to $[6 - 6u(t)]$ A.



Step-by-step solution

Step 1 of 5

Refer to Figure 7.129 in the textbook.

Consider the properties of step input.

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

The expression for the current source is,

$$\begin{aligned} i_{in}(t) &= 6 - 6u(t) \\ &= \begin{cases} 6 & t < 0 \\ 6 - 6 & t > 0 \end{cases} \\ &= \begin{cases} 6 \text{ A} & t < 0 \\ 0 \text{ A} & t > 0 \end{cases} \end{aligned}$$

Step 2 of 5

At $t = 0^-$, the inductor acts as short circuit.

Draw the circuit at $t = 0^-$.

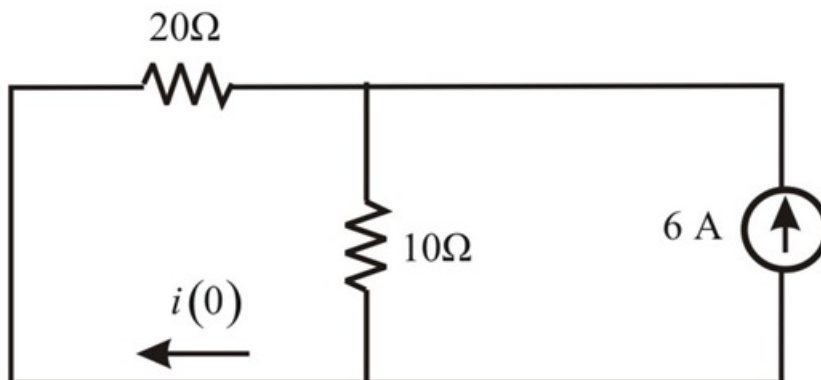


Figure 1

Step 3 of 5

Apply current division rule to calculate the initial current through the inductor.

$$\begin{aligned}i_L(0) &= -\left(\frac{10}{20+10}\right)(6) \\&= -\left(\frac{10}{30}\right)(6) \\&= -2 \text{ A}\end{aligned}$$

For $t > 0$, the circuit becomes source free. So, the final current through the inductor is zero.

$$i_L(\infty) = 0 \text{ A}$$

Draw the circuit to calculate the equivalent resistance for $t > 0$.

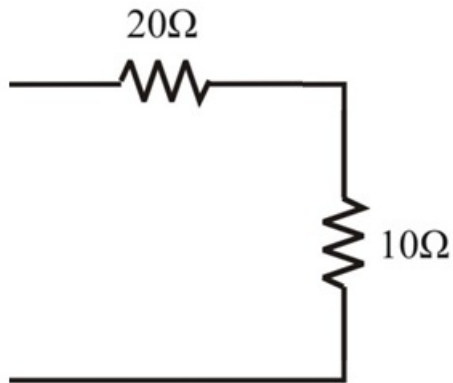


Figure 2

The equivalent resistance for $t > 0$ is,

$$\begin{aligned}R_{eq} &= 20 + 10 \\&= 30 \Omega\end{aligned}$$

Step 4 of 5

Calculate the value of time constant, τ .

$$\tau = \frac{L}{R_{eq}}$$

Substitute 30Ω for R_{eq} and 10 H for L .

$$\begin{aligned}\tau &= \frac{10}{30} \\&= \frac{1}{3} \text{ sec}\end{aligned}$$

Step 5 of 5

Calculate the expression for the inductor current, $i_L(t)$.

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]e^{-\frac{t}{\tau}}$$

Substitute -2 A for $i_L(0)$, 0 A for $i_L(\infty)$ and $\frac{1}{3}$ sec for τ .

$$\begin{aligned} i_L(t) &= 0 + (-2 - 0)e^{-\left(\frac{t}{\frac{1}{3}}\right)} \\ &= -2e^{-3t} \text{ A} \end{aligned}$$

Thus, the expression for the inductor current, $i_L(t)$ is $-2e^{-3t}$ A.

The total energy dissipated by the circuit is equal to the energy stored in the inductor.

Calculate the total energy dissipated by the circuit.

$$W_L = \frac{1}{2}Li^2$$

Substitute -2 A for $i_L(0)$ and 10 H for L .

$$\begin{aligned} W_L &= \frac{1}{2} \times 10 \times (-2)^2 \\ &= \frac{1}{2} \times 10 \times 4 \\ &= 20 \text{ J} \end{aligned}$$

Thus, the total energy dissipated by the circuit is 20 J.